

- RAG Learning

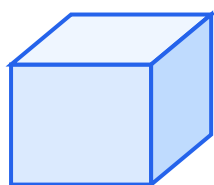
4.4 Mensuration: Area, Volume & 3D

AQA GCSE Mathematics 8300 — Geometry and measures — spec refs G11–G14, G16, G17 — Papers 1 (non-calculator), 2 and 3 (calculator)

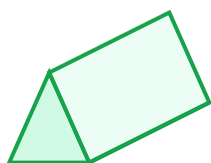
Mensuration is the measurement of length, area and volume. This topic covers the properties of 3D solids, plans and elevations, units of measure, perimeter and area of 2D shapes (including circles and composite shapes), and the volume and surface area of prisms, cylinders, pyramids, cones, spheres and composite solids. Higher questions use the same formulae in harder multi-step problems, such as frustums and solids made from several shapes.

4.4.1 Properties of 3D Solids

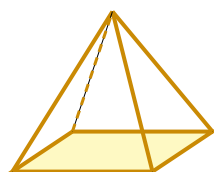
Face — a flat surface. **Surface** — any outside surface, flat or curved. **Edge** — a line where two faces or surfaces meet. **Vertex** (plural vertices) — a corner where edges meet.



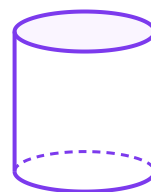
Cuboid



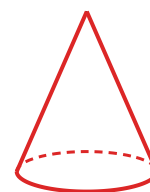
Triangular prism



Square-based pyramid



Cylinder



Cone

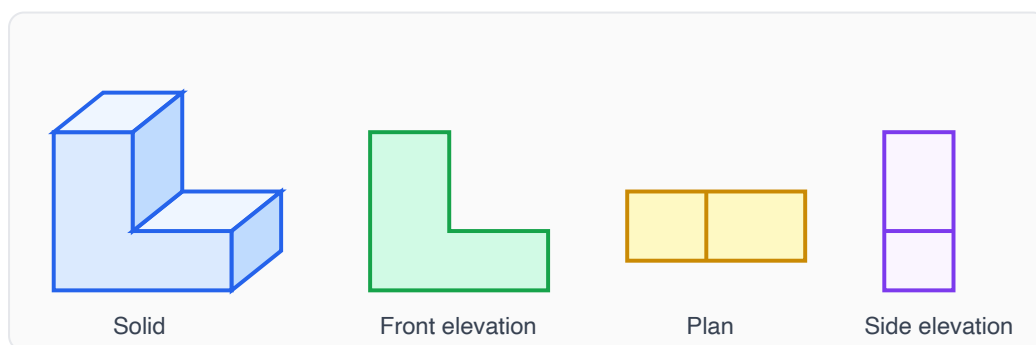
Solid	Faces / surfaces	Edges	Vertices
Cube / cuboid	6 faces	12	8
Triangular prism	5 faces (2 triangles, 3 rectangles)	9	6
Square-based pyramid	5 faces (1 square, 4 triangles)	8	5

Triangular-based pyramid (tetrahedron)	4 faces	6	4
Cylinder	2 flat circular faces + 1 curved surface	2 (curved)	0
Cone	1 flat circular face + 1 curved surface	1 (curved)	1 (apex)
Sphere	1 curved surface	0	0

A **prism** has the same cross-section all the way through. A **pyramid** has a polygon base and triangular faces meeting at an apex. For any polyhedron (flat faces only), Euler's formula $F + V - E = 2$ is a quick check: a cuboid gives $6 + 8 - 12 = 2$.

4.4.2 Plans and Elevations

Plan — the view from directly above. **Front elevation** — the view from the front. **Side elevation** — the view from the side. Each is a 2D drawing showing the true lengths of edges seen from that direction.



Draw a line on a plan or elevation wherever there is a change of depth or height (an edge you can see). On the plan above, the line shows where the tall part meets the low part.

Worked Example: A cuboid 5 cm long, 3 cm wide and 2 cm high rests on a 5 cm by 3 cm face. Describe its plan and front elevation (viewed from the 5 cm side).

Plan: a **rectangle 5 cm by 3 cm**. Front elevation: a **rectangle 5 cm by 2 cm**.

Exam Tip: When constructing plans and elevations on squared paper, use a ruler and keep lengths accurate. Check that the plan's width matches the front elevation's width, and its depth matches the side elevation's width.

4.4.3 Units of Measure

Length, area and volume use different units. Area units are **squared** and volume units are **cubed**, so the conversion factor is squared or cubed too.

Length	Area	Volume / capacity
1 cm = 10 mm	$1 \text{ cm}^2 = 10^2 = 100 \text{ mm}^2$	$1 \text{ cm}^3 = 10^3 = 1000 \text{ mm}^3$
1 m = 100 cm	$1 \text{ m}^2 = 100^2 = 10\,000 \text{ cm}^2$	$1 \text{ m}^3 = 100^3 = 1\,000\,000 \text{ cm}^3$
1 km = 1000 m	$1 \text{ km}^2 = 1\,000\,000 \text{ m}^2$	1 litre = 1000 cm ³ 1 ml = 1 cm ³
		$1 \text{ m}^3 = 1000 \text{ litres}$

Other standard measures: 1 kg = 1000 g, 1 tonne = 1000 kg; 1 hour = 60 minutes; £1 = 100p.

Worked Example: (a) Convert 3.5 m² to cm². (b) Convert 45 000 cm² to m². (c) A tank holds 60 000 cm³. How many litres is this?

(a) $3.5 \times 10\,000 = \mathbf{35\,000 \text{ cm}^2}$

(b) $45\,000 \div 10\,000 = \mathbf{4.5 \text{ m}^2}$

(c) $60\,000 \div 1000 = \mathbf{60 \text{ litres}}$

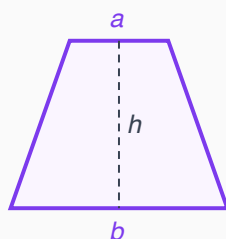
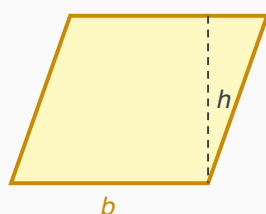
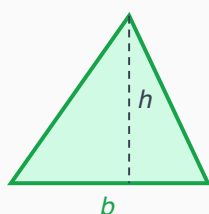
Exam Tip: A classic mistake is converting 1 m² to 100 cm². Picture a 1 m by 1 m square: it is

100 cm by 100 cm, so it contains 10 000 cm².

4.4.4 Perimeter and Area of 2D Shapes

Perimeter — the total distance around the outside of a shape (units: cm, m). **Area** — the amount of flat space inside it (units: cm², m²).

Shape	Area
Rectangle	$A = lw$
Triangle	$A = \frac{1}{2}bh$ (h = perpendicular height)
Parallelogram	$A = bh$ (h = perpendicular height)
Trapezium	$A = \frac{1}{2}(a + b)h$ (a, b = parallel sides)



Worked Example: A trapezium has parallel sides 7 cm and 11 cm, 6 cm apart. Find its area.

$$A = \frac{1}{2}(7 + 11) \times 6 = \frac{1}{2} \times 18 \times 6 = \mathbf{54 \text{ cm}^2}$$

Circles

Circumference: $C = 2\pi r = \pi d$ Area: $A = \pi r^2$

Worked Example: A circle has radius 7 cm. Find its circumference and area to 1 d.p.

$$C = 2 \times \pi \times 7 = 43.98 \dots \approx \mathbf{44.0 \text{ cm}}$$

$$A = \pi \times 7^2 = 153.93 \dots \approx \mathbf{153.9 \text{ cm}^2}$$

Worked Example: Find the perimeter and area of a semicircle of diameter 10 cm, in terms of π .

Curved part = $\frac{1}{2} \times \pi \times 10 = 5\pi$; perimeter = $5\pi + 10 \text{ cm}$ (don't forget the straight edge)

$$\text{Area} = \frac{1}{2} \times \pi \times 5^2 = \mathbf{12.5\pi \text{ cm}^2}$$

In terms of π means leave π as a symbol, e.g. $36\pi \text{ cm}^2$. This is exact and is common on the non-calculator paper.

Composite Shapes

Split the shape into rectangles, triangles, trapezia and parts of circles. Add areas of pieces that are joined; subtract areas of pieces that are cut out.

Worked Example: A 12 cm by 8 cm rectangle has a quarter circle of radius 8 cm, centred at one corner, cut out. Find the remaining area to 1 d.p.

$$\text{Rectangle} = 12 \times 8 = 96 \text{ cm}^2$$

$$\text{Quarter circle} = \frac{1}{4} \times \pi \times 8^2 = 16\pi = 50.26 \dots \text{ cm}^2$$

$$\text{Remaining} = 96 - 16\pi = 45.73 \dots \approx \mathbf{45.7 \text{ cm}^2}$$

Problems on Coordinate Axes

Read lengths from coordinates: horizontal lengths are differences in x , vertical lengths are differences in y .

Worked Example: A triangle has vertices A(1, 2), B(7, 2) and C(7, 6). Find its area.

AB is horizontal: $7 - 1 = 6$. BC is vertical: $6 - 2 = 4$. Angle B is a right angle.

$$\text{Area} = \frac{1}{2} \times 6 \times 4 = \mathbf{12 \text{ units}^2}$$

4.4.5 Volume of Prisms and Cylinders

Prism: $V = \text{area of cross-section} \times \text{length}$

Cuboid: $V = lwh$ **Cylinder:** $V = \pi r^2 h$

Worked Example: A triangular prism is 15 cm long. Its cross-section is a triangle with base 6 cm and height 4 cm. Find its volume.

$$\text{Cross-section} = \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2; \quad V = 12 \times 15 = \mathbf{180 \text{ cm}^3}$$

Worked Example: A cylinder has diameter 10 cm and height 14 cm. Find its volume to 3 s.f.

$$r = 5; \quad V = \pi \times 5^2 \times 14 = 350\pi = 1099.5 \dots \approx \mathbf{1100 \text{ cm}^3}$$

Worked Example: A cuboid tank 80 cm by 50 cm by 40 cm holds water 30 cm deep. How many more litres are needed to fill it?

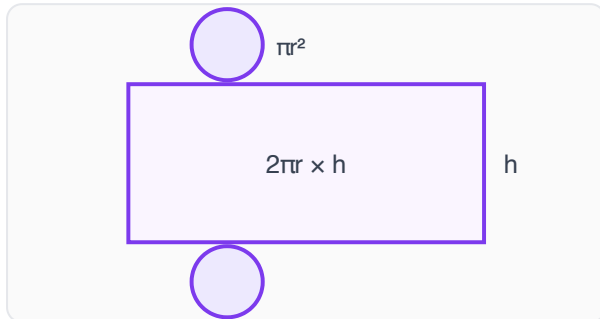
$$\text{Empty depth} = 10 \text{ cm, so volume needed} = 80 \times 50 \times 10 = 40\,000 \text{ cm}^3 = \mathbf{40 \text{ litres}}$$

4.4.6 Surface Area of Prisms and Cylinders

Surface area is the total area of all the outside faces. Sketching the **net** helps you not miss a face.

Cuboid: $SA = 2(lw + lh + wh)$

Cylinder: curved surface = $2\pi rh$; total (closed) = $2\pi r^2 + 2\pi rh$



Worked Example: Find the total surface area of a closed cylinder with radius 5 cm and height 12 cm, in terms of π .

Ends: $2 \times \pi \times 5^2 = 50\pi$. Curved: $2 \times \pi \times 5 \times 12 = 120\pi$.

Total = **$170\pi \text{ cm}^2$**

Worked Example: A prism 10 cm long has a right-angled triangle cross-section with sides 3, 4 and 5 cm. Find its surface area.

Two triangles: $2 \times \frac{1}{2} \times 3 \times 4 = 12$. Three rectangles: $(3 + 4 + 5) \times 10 = 120$.

Total = **132 cm^2**

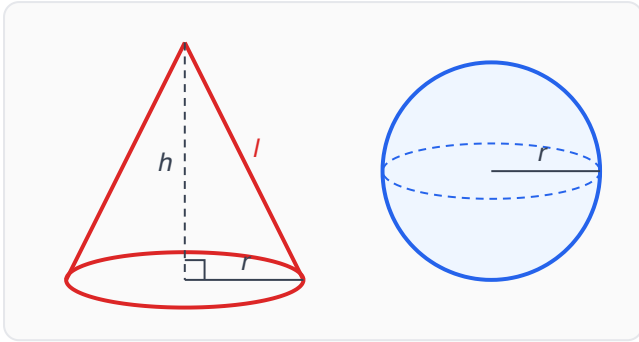
4.4.7 Pyramids, Cones and Spheres

Pyramid: $V = \frac{1}{3} \times \text{base area} \times h$

Cone: $V = \frac{1}{3} \pi r^2 h$; curved surface area = $\pi r l$ (l = slant height)

Sphere: $V = \frac{4}{3} \pi r^3$; surface area = $4\pi r^2$

Hemisphere: $V = \frac{2}{3} \pi r^3$; curved surface = $2\pi r^2$



The cone, sphere and curved-surface formulae are usually given in the question when needed, but you must be able to use them. The slant height l , perpendicular height h and radius r of a cone form a right-angled triangle: $l^2 = r^2 + h^2$.

Worked Example: A square-based pyramid has volume 192 cm^3 and height 9 cm . Find the length of the base.

$$\frac{1}{3} \times x^2 \times 9 = 192 \Rightarrow 3x^2 = 192 \Rightarrow x^2 = 64 \Rightarrow x = 8 \text{ cm}$$

Worked Example: A solid cone has base radius 6 cm and height 8 cm . Find its total surface area in terms of π .

$$\text{Slant height } l = \sqrt{6^2 + 8^2} = 10 \text{ cm}$$

$$\text{Curved surface} = \pi \times 6 \times 10 = 60\pi; \text{ base} = \pi \times 6^2 = 36\pi$$

$$\text{Total} = 96\pi \text{ cm}^2$$

Worked Example: A sphere has radius 4.5 cm . Find its volume to 3 s.f.

$$V = \frac{4}{3} \times \pi \times 4.5^3 = 121.5\pi = 381.7 \dots \approx 382 \text{ cm}^3$$

Exam Tip: For a cone's volume use the **perpendicular** height h ; for its curved surface use the **slant** height l . If you are given the wrong one, use Pythagoras first.

4.4.8 Composite Solids and Frustums Higher

A composite solid is made from two or more standard solids. For **volume**, add (or subtract) the volumes of the parts. For **surface area**, add only the surfaces on the outside — leave out any faces where the parts are joined.

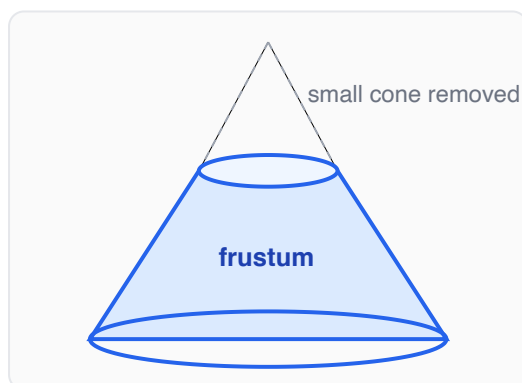
Worked Example: A silo is a cylinder (radius 3 m, height 10 m) with a hemisphere of radius 3 m on top. Find its volume to 3 s.f.

$$\text{Cylinder} = \pi \times 3^2 \times 10 = 90\pi; \text{ hemisphere} = \frac{2}{3} \times \pi \times 3^3 = 18\pi$$

$$\text{Total} = 108\pi = 339.2 \dots \approx \mathbf{339 \text{ m}^3}$$

A **frustum** is the solid left when the top of a cone (or pyramid) is cut off by a plane parallel to the base.

$$V_{\text{frustum}} = V_{\text{large cone}} - V_{\text{small cone}}$$



Worked Example: A frustum is made by removing a cone of height 6 cm and radius 3 cm from the top of a cone of height 18 cm and radius 9 cm. Find its volume to 3 s.f.

$$\text{Large cone} = \frac{1}{3} \times \pi \times 9^2 \times 18 = 486\pi$$

$$\text{Small cone} = \frac{1}{3} \times \pi \times 3^2 \times 6 = 18\pi$$

$$\text{Frustum} = 486\pi - 18\pi = 468\pi = 1470.2 \dots \approx \mathbf{1470 \text{ cm}^3}$$

If the question gives only the frustum's two radii and depth, use **similar triangles** to find the height of the full cone. For a bucket with radii 15 cm and 10 cm and depth 20 cm: $\frac{H-20}{H} = \frac{10}{15}$ gives $H = 60$ cm, so the small cone has height 40 cm.

Equal Volumes and Melting/Recasting

Worked Example: A solid metal sphere of radius 6 cm is melted and recast into cones of radius 3 cm and height 4 cm. How many cones can be made?

$$\text{Sphere} = \frac{4}{3} \times \pi \times 6^3 = 288\pi; \text{cone} = \frac{1}{3} \times \pi \times 3^2 \times 4 = 12\pi$$

$$288\pi \div 12\pi = \mathbf{24 \text{ cones}}$$

Worked Example: A cylinder of radius 6 cm holds water 15 cm deep. A sphere of radius 4.5 cm is dropped in and sinks. Find the new depth.

$$\text{Sphere volume} = 121.5\pi; \text{rise} = 121.5\pi \div (\pi \times 6^2) = 3.375 \text{ cm}$$

$$\text{New depth} = 15 + 3.375 = \mathbf{18.4 \text{ cm}} \text{ (to 3 s.f.)}$$

Exam Tip: Keep answers in terms of π through the working — in "how many" questions the π often cancels. When the answer must be a whole number of items, round *down* for "how many can be made" and *up* for "how many tins/boxes are needed".

Summary of Formulae

Shape	Volume	Surface area
Cuboid	lwh	$2(lw + lh + wh)$
Prism	cross-section $\times$ length	sum of all faces
Cylinder	$\pi r^2 h$	$2\pi r^2 + 2\pi rh$
Pyramid	$\frac{1}{3} \times \text{base} \times h$	base + triangular faces

Cone	$\frac{1}{3}\pi r^2 h$	$\pi r l + \pi r^2$
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$
Hemisphere (solid)	$\frac{2}{3}\pi r^3$	$3\pi r^2$